Microcontroller

File: microcontroller.kicad_sch

Sensor

File: sensor.kicad_sch

Power

File: power.kicad_sch

Connector

File: connector.kicad_sch

Microcontroller

Aksantara ITB

Sheet: /

File: SvayamV3_IIK.kicad_sch

Title: Svayamkriya Vibhava

Size: A4	Date:	**Rev:**
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+3V3_FMU

C1 1u
C2 100n
C3 4u7
C4 1u
C5 100n
C6 100n
C7 100n
C8 100n
C9 100n
C10 100n
C11 100n
C12 100n
C14 100n
C18 100n
C20 100n
C21 100n
C22 100n
C23 100n
C24 100n

PWR_FLAG

13 VDD → 100nF x 13: 4.7uF
VREF+ VDDA+ → 100nF; 1uF
VDD33_USB → 100nF

R5 10k

TS-1010A-B
SW2
C15 100n
NRST

+3V3_FMU
SW1
SK-3296S-01-L3
R4 10k
+3V3_FMU

Y1
X50328MSB2GI 8 MHz
C13 33p
C16 33p

+5V_IN
D1 PWR
D2 BOOT
D3 STAT
R1 330
R2 1k5
R3 1k5
LED_BOOT
LED_ST

PC13 / PC14:
open-drain output
Active Low

J1 NRST
C6 PDR_ON
D6 BOOT0
P1 VREF+
N1 VREF-
VBAT
C1 C5 C7 C8 C9 F3 G3 G13 J12 J13 K4 N8 N9 N10 H13 R1 VDD
VDD33_USB VDDA

E14 PI0 BARO2_CS_N
D14 PI1 SPI2_SCK BARO2_SPI2_SCK
C14 PI2 TIM8_CH4 TIM8_CH4
C13 PI3 SPI2_MOSI BARO2_SPI2_MOSI
D4 PI4
C4 PI5
C3 PI6
C2 PI7
D2 PI8
D3 PI9
E3 PI10
E4 PI11

G1 RCC_OSC_IN
H1 RCC_OSC_OUT
F4 PH2
G4 PH3
H4 PH4
J4 PH5
M11 PH6
N12 PH7
M12 PH8
M13 PH9
L13 PH10 TIM5_CH1
L12 PH11 TIM5_CH2
K12 PH12
E12 PH13 UART4_TX
E13 PH14 UART4_RX
D13 PH15

N7 PG0
M7 PG1
L15 PG2
K15 PG3 SAFETY_IN
K14 PG4
K13 PG5
J15 PG6 SAFETY_LED
J14 PG7
H14 PG8
C10 PG9 USART6_RX
B10 PG10 CS_BMI088_ACCEL
B9 PG11 CS_BMI088_GYRO
B8 PG12 SPI6_MISO
A8 PG13 SPI6_SCK
A7 PG14 SPI6_MOSI
B7 PG15

E2 PF0
H3 PF1 PERI_EN
H2 PF2 PWR_SEL
J2 PF3 ADC3_INP5
J3 PF4
K3 PF5 CS_EXT1
K2 PF6 ADC3_INP8
K1 PF7 SPI5_SCK
L3 PF8 SPI5_MISO
L2 PF9 SPI5_MOSI
L1 PF10 CS_EXT2
R6 PF11
P6 PF12
N6 PF13
R7 PF14 I2C4_SCL
P7 PF15 I2C4_SDA

F13 VCAP
M10 VCAP

C17 2u2
C19 2u2

D5 VSS
M1 VSSA

U1
STM32H743IIK6

N3 PA0 TIM2_CH1
N2 PA1 TIM2_CH2
P2 PA2 TIM2_CH3
R2 PA3 PERI_OC
N4 PA4
P4 PA5
P3 PA6
R3 PA7 TIM3_CH2
F15 PB0 TIM1_CH1
E15 PB1 TIM1_CH2
D15 PB10 TIM1_CH3
C15 PB11
B15 PB12 USB_D-
A15 PB13 USB_D+
A14 PB14 SWDIO
A13 PB15 SWCLK
PA15

R5 TIM3_CH3
R4 PB0 TIM3_CH4
M6 PB1
A10 PB2
A9 PB3 SPI1_SCK
A6 PB4 SPI1_MISO
B6 PB5 CS_BMI270
B5 PB6 I2C1_SCL
A5 PB7 I2C1_SDA
B4 PB8 FDCAN1_RX
R12 PB9 FDCAN1_TX
R13 PB10 USART3_TX
P12 PB11 USART3_RX
P13 PB12 FDCAN2_RX
R14 PB13 FDCAN2_TX
R15 PB14 USART1_TX
USART1_RX

M2 PB15 ADC1_INP10
M3 PC0 ADC1_INP11
M4 PC1 SPI2_MISO BARO2_SPI2_MISO
M5 PC2_C
N5 PC3 ADC2_INP4
P5 PC4 ADC2_INP8
H15 PC5 TIM8_CH1
G15 PC6 TIM8_CH2
G14 PC7 SDMMC1_D0 SD_DAT0
F14 PC8 SDMMC1_D1 SD_DAT1
B14 PC9 SDMMC1_D2 SD_DAT2
B13 PC10 SDMMC1_D3 SD_DAT3
A12 PC11 SDMMC1_CK SD_CLK
D1 PC12 LED_ST
E1 PC13 LED_BOOT
F1 PC14
PC15

B12 PD0 GPS_OC
C12 PD1 TELEM1_OC
D12 PD2 SDMMC1_CMD SD_CMD
D11 PD3 USART2_CTS USART2_CTS
D10 PD4 USART2_RTS USART2_RTS
C11 PD5 USART2_TX USART2_TX
B11 PD6 USART2_RX USART2_RX
A11 PD7 SPI1_MOSI SPI1_MOSI
P15 PD8 TELEM2_OC
P14 PD9 SERIAL4_OC
N15 PD10 GPIO
N14 PD11 USART3_CTS USART3_CTS
N13 PD12 USART3_RTS USART3_RTS
M15 PD13 TIM4_CH2
M14 PD14 TIM4_CH3
L14 PD15 TIM4_CH4 ALARM

A4 PE0
A3 PE1
A2 PE2 SPI4_SCK
A1 PE3
B1 PE4
B2 PE5 SPI4_MISO
B3 PE6 SPI4_MOSI
R8 PE7 UART7_RX
P8 PE8 UART7_TX
P9 PE9 UART7_RTS
R9 PE10 UART7_CTS
P10 PE11 SERIAL5_OC
R10 PE12 I2C_OC
N11 PE13 CAN_OC
P11 PE14 CAN2_OC
R11 PE15 SPI_OC

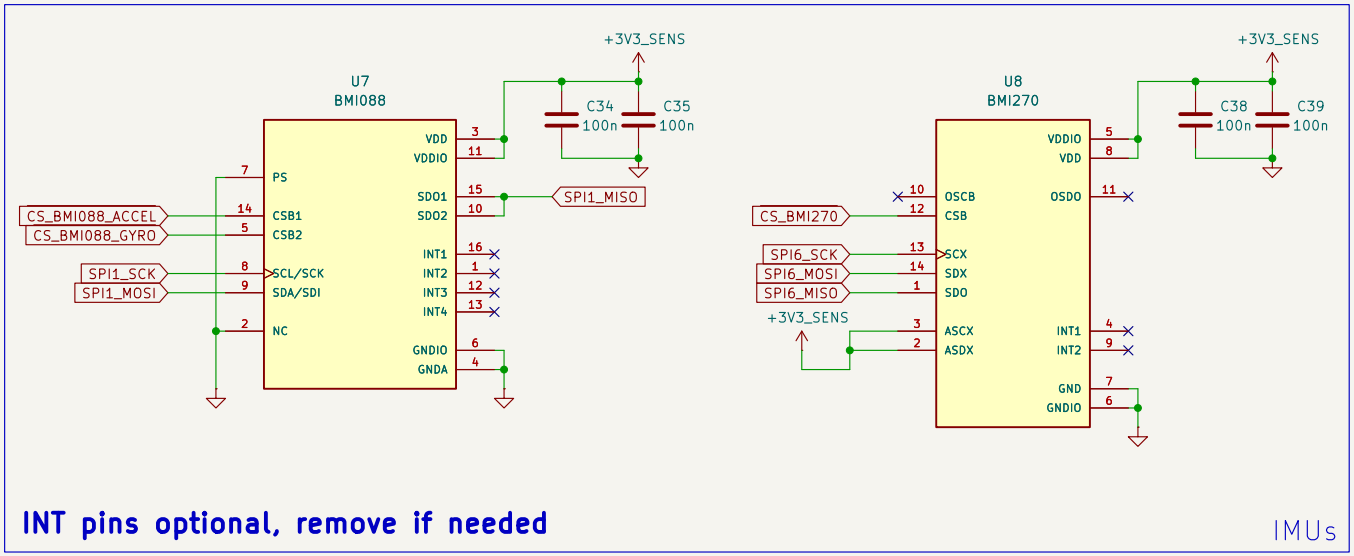
+5V_PERIPH
C27 100n
U3 TJA1051TK-3
TXD RXD VCC CANH CANL VIO S GND GND
R13 120
CANH CANL
+3V3_FMU
C25 100n

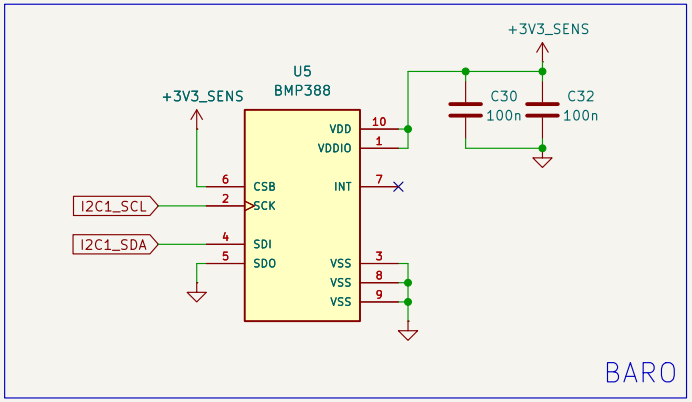
+5V_PERIPH
C29 100n
U4 TJA1051TK-3
R14 120
CANH2 CANL2
+3V3_FMU
C28 100n

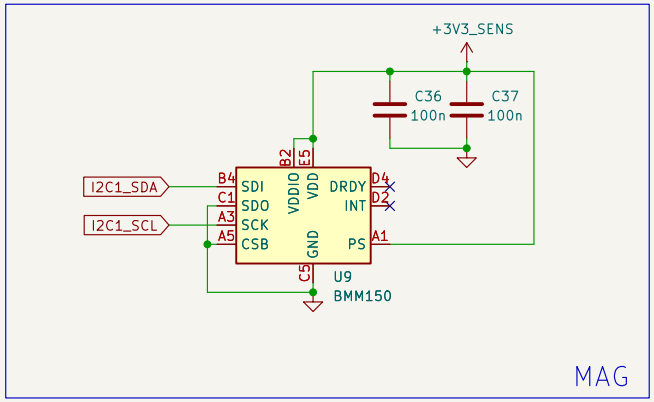
+3V3_FMU
R7 1k5
R8 1k5
R10 1k5
R11 1k5
I2C4_SCL I2C4_SDA I2C1_SDA I2C1_SCL
I2C Pull Up Res

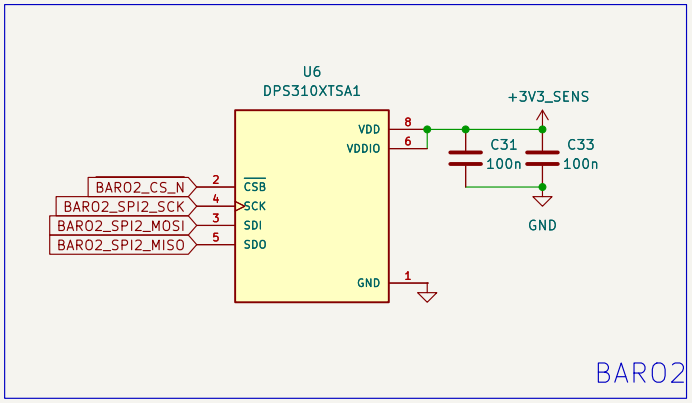
+3V3_FMU
R6 10k
C26 100n
U2 FM25CL64B
CS SCK SI SO VDD VSS WP HOLD
R9 10k
R12 10k

Aksantara ITB
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File: microcontroller.kicad_sch
Title: Svayamkriya Vibhava
Size: A3 | Date: | Rev:
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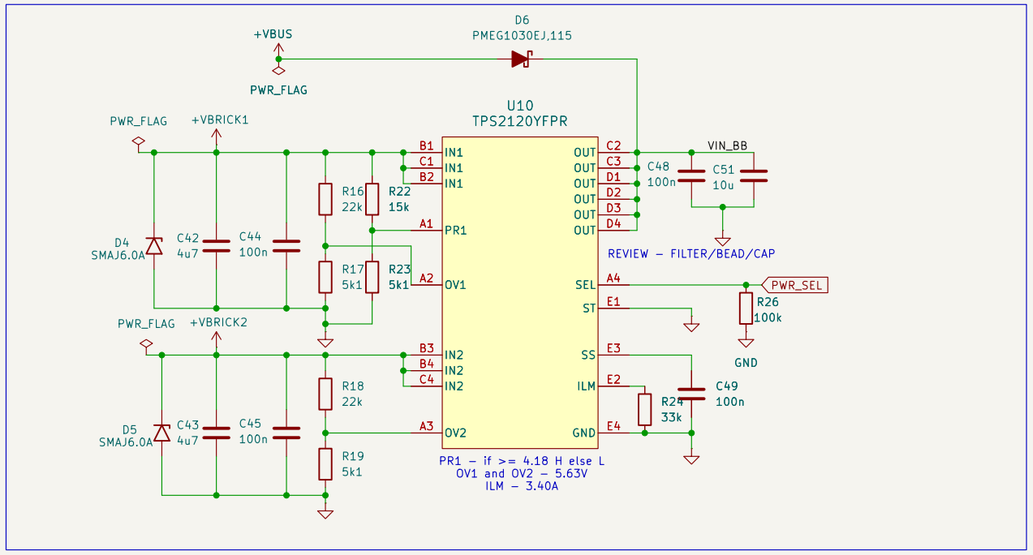


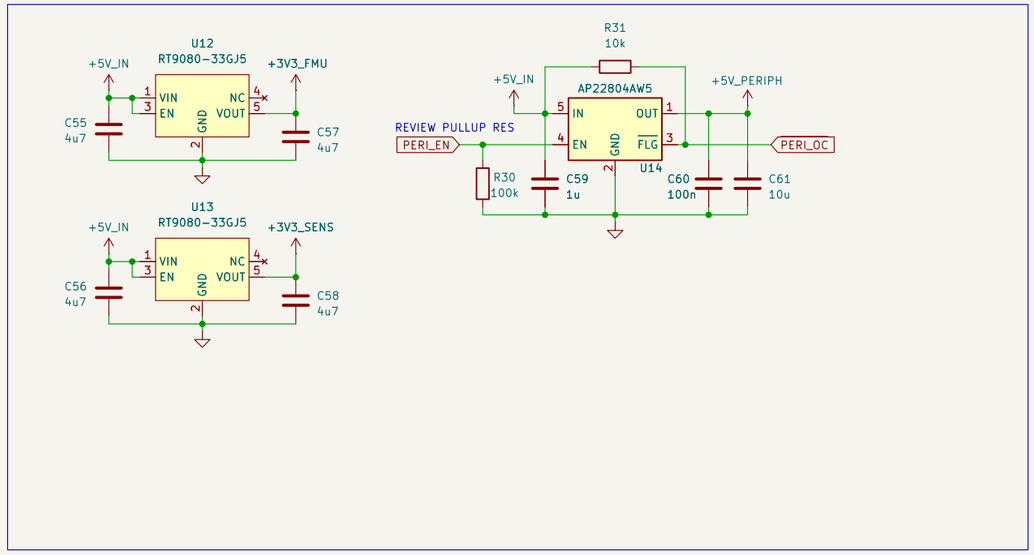


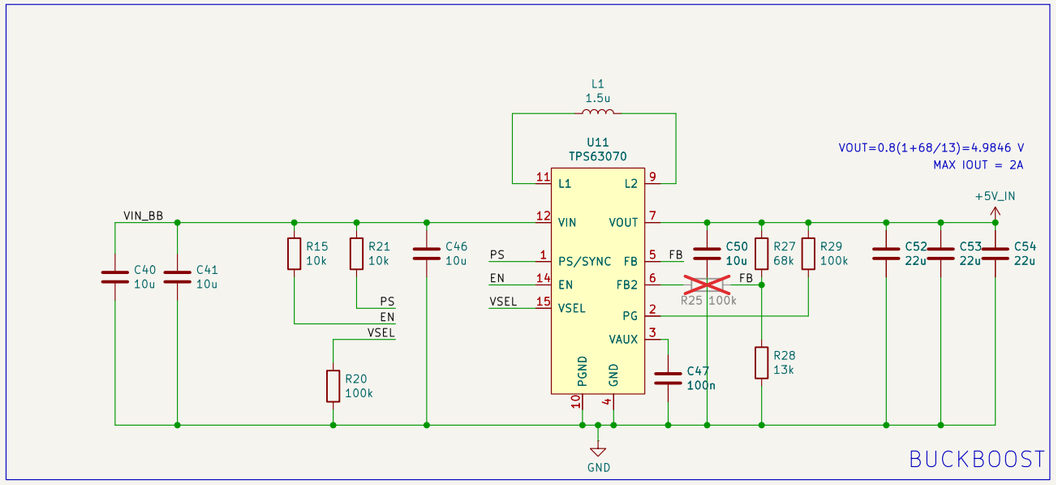




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Sheet: /Sensor/		
File: sensor.kicad_sch		
Title: Svayamkriya Vibhava		
Size: A4	Date:	**Rev:**
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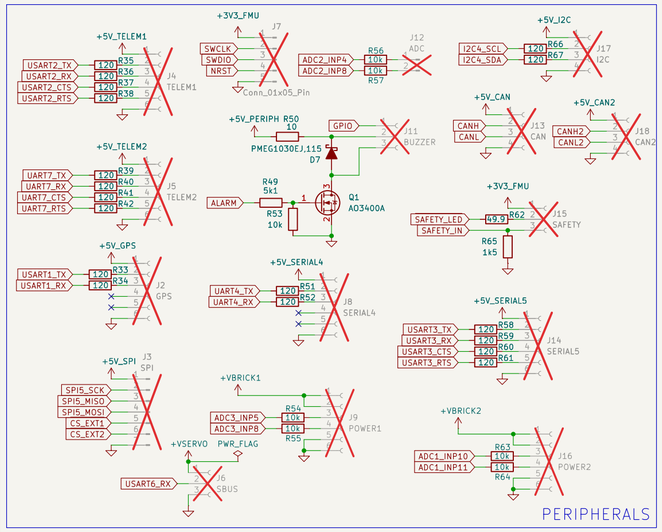
Aksantara ITB

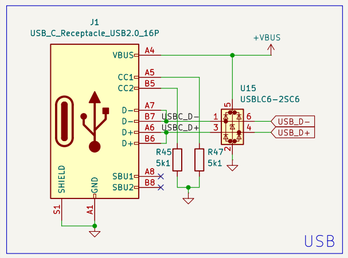
Sheet: /Power/

File: power.kicad_sch

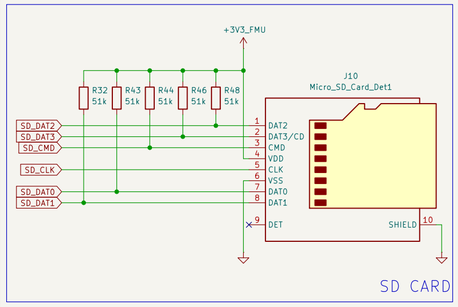
Title: Svayamkriya Vibhava

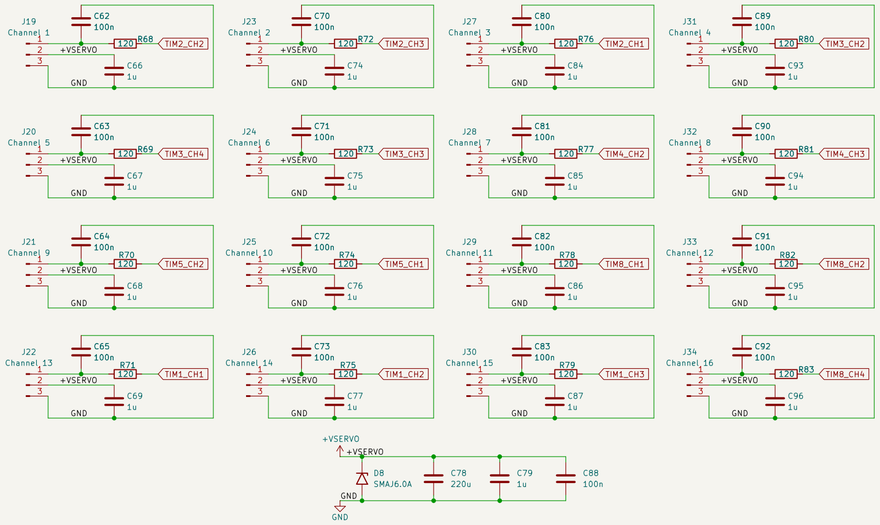
Size: A3	Date:	Rev:
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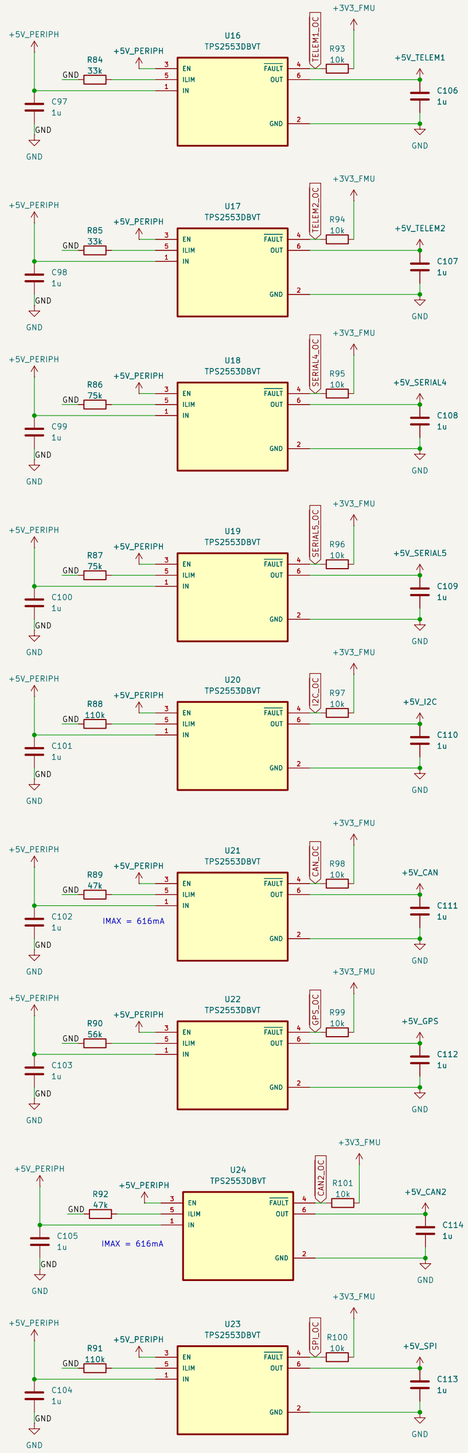












Aksantara ITB
Sheet: /Connector/
File: connector.kicad_sch
Title: Svayamkriya Vibhava
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KiCad E.D.A. 10.0.6 | Id: 5/5